

# Example



- A sample of sand has the following properties:
  - Moist mass = 625.2 g
  - Dry mass = 589.9 g
  - Absorption = 1.6%
- Determine: (a) total moisture content, and (b) free moisture content.

# Solution



(a) Mass of water =  $625.2 - 589.9 = 35.3$  g

Total moisture content  
=  $(35.3/589.9) \times 100 \%$   
= **6.0 %**

$$M = \frac{W_m - W_{OD}}{W_{OD}}$$

(b) Free moisture  
=  $6.0 - 1.6$   
= **4.4 %**

**Percent free moisture = M - A**



# Specific Gravity

- ❑ Knowing density of aggregate is required in concrete **mix design** to establish weight- volume relationships
- ❑ Density is expressed as specific gravity
- ❑ Specific gravity (SG) is a dimensionless ratio relating density of aggregate to that of water

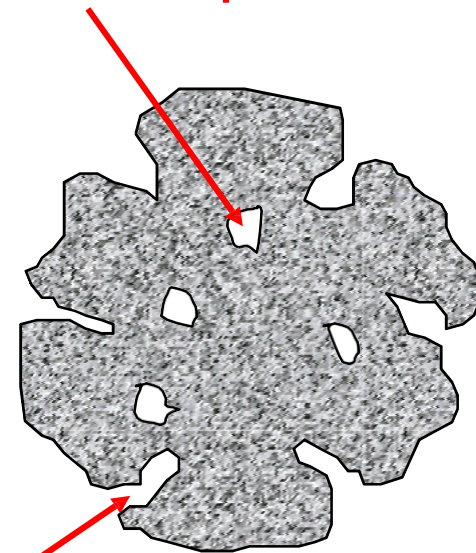
$$SG = \frac{\text{density of solid}}{\text{density of water}}$$

Density of water @ 4°C is 1000 kg/m<sup>3</sup>

# Effects of Voids in Aggregates

- Permeable pores in aggregates create multiple definitions of specific gravity
  - Apparent, ASG
  - Bulk-dry,  $BSG_{OD}$
  - Bulk-saturated surface dry,  $BSG_{SSD}$

Impermeable pores



Permeable pores

# Apparent Specific Gravity

$$ASG = \frac{\text{Dry weight of Agg.}}{\text{Vol. of Agg. + impermeable pores}} \cdot \frac{1}{\rho_w}$$

$$= \frac{W_{OD}}{(V_s + V_i)\rho_w}$$

Solid

Impermeable pore

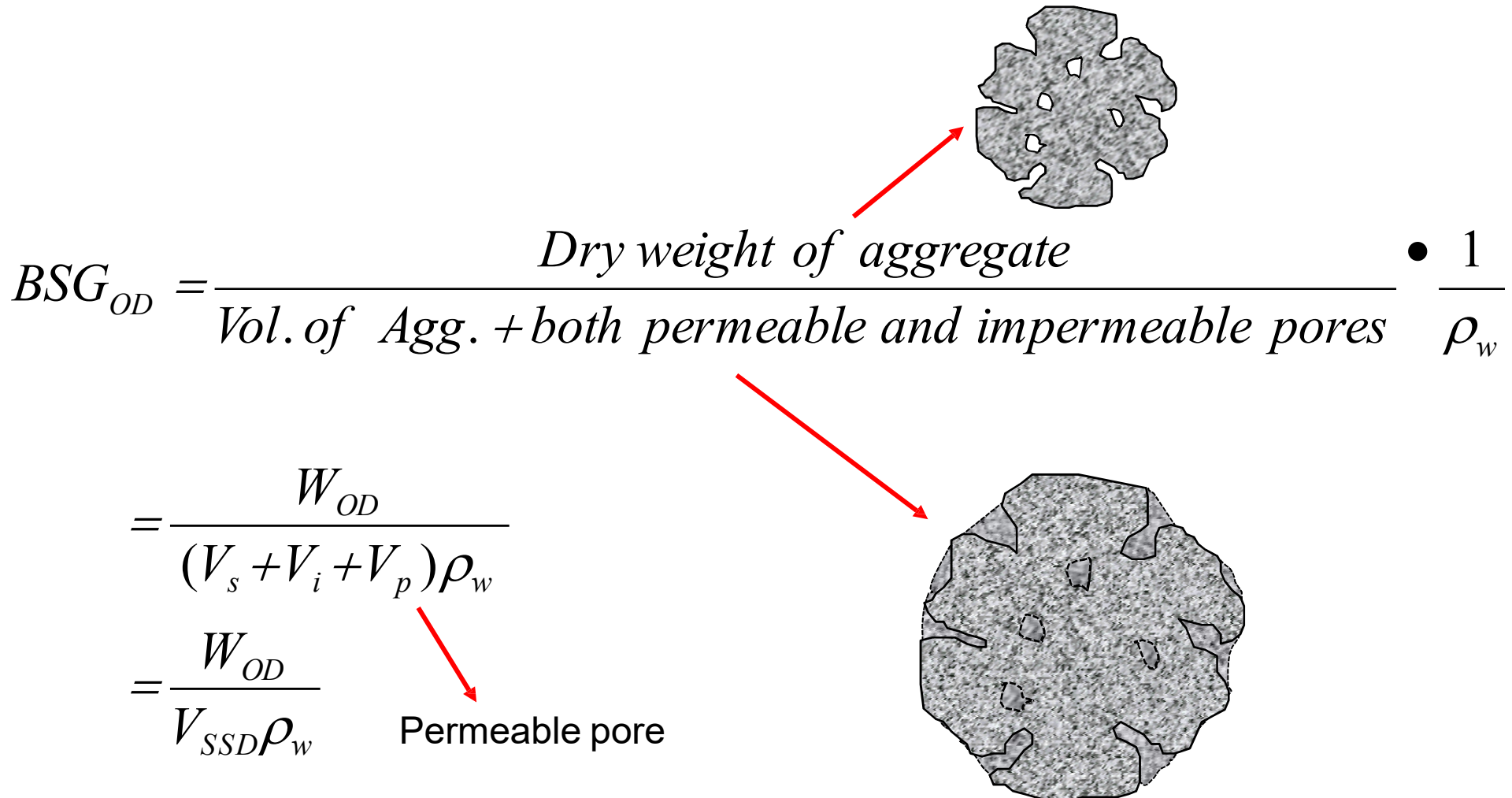
# Bulk Specific Gravity, Oven Dry

$$BSG_{OD} = \frac{\text{Dry weight of aggregate}}{\text{Vol. of Agg. + both permeable and impermeable pores}} \cdot \frac{1}{\rho_w}$$

$$= \frac{W_{OD}}{(V_s + V_i + V_p)\rho_w}$$

$$= \frac{W_{OD}}{V_{SSD}\rho_w}$$

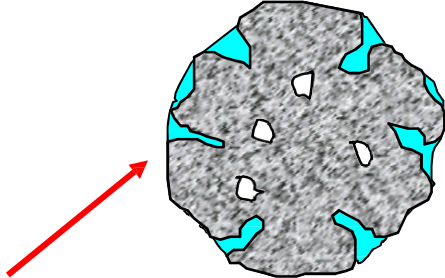
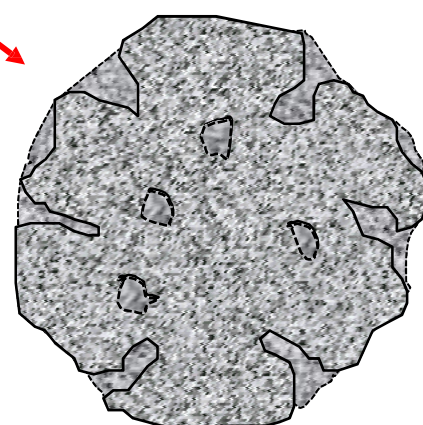
Permeable pore



# Bulk Specific Gravity, SSD

$$BSG_{SSD} = \frac{\text{SSD weight of aggregate}}{\text{Vol. of Agg. + both permeable and impermeable pores}} \bullet \frac{1}{\rho_w}$$

$$= \frac{W_{OD} + W_p}{(V_s + V_i + V_p)\rho_w}$$

$$= \frac{W_{SSD}}{V_{SSD}\rho_w}$$





# BSG<sub>SSD</sub> for Concrete Mix Design

- BSG<sub>SSD</sub> is used as the reference state for mix design purpose
  - ❖ Water in permeable pores of aggregate does not participate in cement hydration
  - ❖ Effective volume that SSD aggregate occupies in concrete include permeable pores
  - ❖ If non-SSD aggregates are used in producing concrete based on BSG<sub>SSD</sub> concrete mix design, the resulting water-to-cement ratio is different than the intended design
  - ❖ If non-SSD aggregates are expected to be used, corrections should be taken into account

